JJJ-W-141C
May 21, 1975
SUPERSEDING
JJJ-W-141B
August 15, 1957

FEDERAL SPECIFICATION

WAX, CARNAUBA

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers refined and crude carnauba wax.

1.2 Classification. Carnauba wax shall be of the following types and grades as specified (see 6.2).

Type I. Yellow, which includes the following:

Grade A - Flor.
Prime or No. 1 Yellow.

Grade B - Medium or No. 2 Yellow.

Type II. Fatty, which includes the following.

Grade A - Caahype.
Corda Clara.
Light Fatty or No. 2 North Country.

Grade B - Fatty Gray or No. 3 North Country.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-35 - Bags, Textile, Shipping, Burlap, Cotton And Waterproof Laminated.
RR-S-366 - Sieve, Test.

Federal Standard:

FED. STD. NO. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks [illegible] established distribution procedure.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

21 CFR 1 - Federal Food, Drug and Cosmetic Act.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible thereof).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Wax Importers and Refiners Association, Inc. (Amerwax).

Wax Analytical Test Methods.

(Application for copies should be addressed to the American Wax Importers and Refiners Association, Inc., 255 West 34th Street, New York, NY 10001.)

American Society for Testing and Materials (ASTM):

D 5 - Penetration of Bituminous Materials, Test for.
D 71 - Specific Gravity of Solid Pitch and Asphalt (Displacement Method).
D 92 - Flash and Fire Points by Cleveland Open Cup.
D 1342 - Paraffin - Type Hydrocarbons in Carnauba Wax.
D 1747 - Refractive Index of Viscous Materials.
D 1959 - Iodine Value of Drying Oils and Fatty Acids.
D 1965 - Unsaponifiable Matter in Drying Oils, Fatty Acids, and Polymerized Fatty Acids.
E 1 - ASTM Thermometers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. The wax shall be a very hard, brittle, nontacky, lustrous material made directly from the powder obtained from the leaves of the Carnauba Palm (Copernicia Cerifera), and shall meet the requirements as specified in table I and section 3.

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Table I. Requirements

Characteristics	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Melting point, °C	82.5	86.0	82.5	86.0	82.5	86.0	82.5	86.0
Needle penetration at 25°C	0.5	1.0	0.5	1.0	0.5	1.0	0.5	1.0
Flash point, °C	299	----	299	----	299	----	299	----
Specific gravity at 25°C	0.996	0.998	0.996	0.998	0.996	0.998	0.996	0.998
Acid number	2.0	6.0	4.0	10.0	2.0	6.0	4.0	10.0
Ester number	66.0	78.0	66.0	78.0	66.0	78.0	66.0	78.0
Saponification number	72.0	88.0	72.0	88.0	72.0	88.0	72.0	88.0
Iodine number	7.0	14.0	7.0	14.0	7.0	14.0	7.0	14.0
Refractive index at 80°C	1.454	1.454	1.454	1.454	1.454	1.454	1.454	1.454
Unsaponifiable matter, percent	50.0	55.0	50.0	55.0	50.0	55.0	50.0	55.0
Paraffinic hydrocarbon, percent	----	2.0	----	2.0	----	2.0	----	2.0
Resinous matter, percent	----	3.5	----	3.5	----	3.5	----	3.5
Benzene solubility, percent	----	8.0	----	8.0	----	8.0	----	8.0
Insoluble impurities, percent	----	*	----	*	----	1.0	----	2.0
Volatile matter, percent	----	*	----	*	----	2.0	----	2.0

* [illegible]

3.2 Color. Color shall be determined according to 4.18. When color is specified (see 6.2), the wax shall be no darker than the specified color.

3.3 Registration. Carnauba wax shall be registered with the Food and Drug Administration under the Federal Food, Drug, and Cosmetic Act provided it is to be used as a general food additive.

3.4 Workmanship. The carnauba wax furnished in accordance with this specification shall be uniform and free of extraneous matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Certificates of compliance. Where certificates are submitted, the Government reserves the right to check-test such items to determine the validity of the certification.

4.3 Quality conformance inspections.

4.3.1 Quality conformance inspection sampling. Unless otherwise specified, sampling shall be in accordance with MIL-STD-105, at the inspection levels and acceptable quality levels (AQL's) indicated herein:

[illegible]	[illegible]	[illegible]
4.3.2	S-2	2.5
4.3.3	S-2	2.5
4.3.4	S-1	N/A

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4.3.2 Inspection of preparation for delivery. An inspection shall be made to determine conformance to the requirements of section 5. The sample unit for this inspection shall be one filled shipping bag, fully packed, marked, and ready for shipment.

Examine	Defect
Marking	Missing, incorrect, or illegible; not as specified.
Packing	Level not as specified.
Packaging	Level not as specified.
Contents	Net contents less than specified.
Workmanship	Weak or unsound seams; tears, cuts, splits, cracks, or holes; closure incomplete or inadequate; evidence of leakage or seepage from any part of the container; loose or otherwise defective container.

4.3.3 Examination for closure of shipping bag. When shipping bags are required to be in accordance with PPP-B-35, examination for defects in closure shall be in accordance with the appendix of that specification.

4.3.4 Inspection of the end item. An inspection shall be made to determine conformance to the material and workmanship requirements of section 3. The sample unit for this inspection shall be the contents of one filled unit container as specified in 5.1 and 6.2.

Examine	Defect
Material	Not a solid, tacky.
Workmanship	Not uniform, not free from foreign matter.

4.3.5 Testing of the end item. The end item shall be tested for the applicable characteristics as indicated in table I for each lot presented for inspection. A representative composite sample shall be taken by the Government inspector from each inspection lot, placed in a clean, dry container labeled to show in detail all identifying marks of the shipment and the date of sampling. The composite sample shall be taken from not less than ten percent of the packages, but not less than two packages in the shipment. Tests for insoluble impurities and volatile matter are made with the original wax while other tests must be made with clear filtered wax composite. Failure to meet the specified requirements shall be a cause for rejection.

TABLE II. Directions for testing of the end item

Characteristic	Specification reference: Requirement	Specification reference: Test Method	No. of determinations	Results reported: Pass-fail	Results reported: Numerical
Melting point	Table I	4.5	1		0.5°C
Needle penetration	Table I	4.6	1		0.1
Flash point	Table I	4.7	1		1.0°C
Specific gravity	Table I	4.8	1		0.001
Acid number	Table I	4.9	1		0.1
Ester number	Table I	4.9	1		0.1
Saponification number	Table I	4.9	1		0.1
Iodine number	Table I	4.10	1		0.1
Refractive index	Table I	4.11	1		0.001
Unsaponifiable matter	Table I	4.12	1		0.1%
Paraffinic hydrocarbon	Table I	4.13	1		0.1%
Resinous matter	Table I	4.14	1		0.1%
Benzene solubility	Table I	4.15	1		0.1%
Insoluble impurities	Table I	4.16	1		0.1%
Volatile matter	Table I	4.17	1		0.1%
Color	3.2	4.18	1	X	

4.4 Test procedures.

4.4.1 Preparation of sample. The composite sample selected in accordance with 4.3.5 shall be ground to pass entirely through a [illegible] sieve conforming

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to RR-S-366. The sample shall then be thoroughly mixed and quartered down in a tightly stoppered bottle. Portions of this ground sample shall be used for the determination of volatile matter (4.17) and insoluble impurities (4.16).

4.4.1.1 For the remaining tests, a portion of the ground sample prepared as specified in 4.4 shall be melted on a water bath and agitated. While being agitated, the wax shall be poured into shallow pans about one-half inch deep. After cooling, the wax shall be remelted, dehydrated, and filtered through a neutral filter aid.

4.5 Melting point. Melt a 10-gram (g) sample of wax prepared as specified in 4.4.1 in an evaporating dish at the lowest temperature at which the sample is completely melted and clear. Carefully insert the end of a melting point capillary tube into the molten wax to a depth of approximately 1 cm. Remove the tube from the wax while holding a finger over the open end, and wipe clean of excess wax adhering to the outside of the tube. Allow the wax inside of the tube to harden and set aside for 2 hours. Attach the melting point tube containing the wax to a thermometer conforming to ASTM D 1 by means of a rubber band, making sure the wax sample rests alongside the thermometer bulb. The thermometer and melting point tube are then immersed in a modified Thiele Melting Point Tube containing recently boiled water. A small Bunsen burner is lighted and placed under the Thiele tube and heated at a rate of 3°C per minute. The temperature at which the wax rises in the capillary tube is taken as the melting point.

4.6 Needle penetration. The needle penetration of the wax shall be determined in accordance with ASTM D 5.

4.7 Flash point. Melt sufficient amount of the filtered composite wax as specified in 4.4.1 to fill the Cleveland Open Cup to the indicated depth. The flash point of the wax shall be determined in accordance with ASTM D 92.

4.8 Specific gravity. The specific gravity of the wax shall be determined in accordance with ASTM D 71."

4.9 Acid, ester, and saponification numbers.

4.9.1 Apparatus.

(a) Two-250 ml flasks made of alkali resistant glass.
(b) Two-Reflux condensers.
(c) Two-Bunsen burners.

4.9.2 Reagents.

(a) Alcoholic potassium hydroxide (N/2). Standardized.
(b) Aqueous hydrochloric acid (N/2). Standardized.
(c) Neutralized isopropanol and toluol (5:4).

4.9.3 Procedure. Weigh accurately 3 g ± 1 mg of the composite wax and transfer to a 250-ml flask made of alkali resistant glass. Add 50 ml of neutralized isopropanol and toluol reagent and 5 to 15 drops of phenolphthalein indicator solution and dissolve under a reflux condenser. When completely dissolved, titrate with N/2 alcoholic potassium hydroxide to a permanent pink color, which will not disappear after very gently boiling the mixture for one-half minute.

Calculation of acid number.

$$\text{Acid number} = \frac{AN \times 56.1}{C}$$

where:

A = Milliliters of alkali solution required for titration of the sample.
N = Normality of the alkali solution.
C = Grams of sample used.

To the solution resulting from the determination of the acid number, add exactly 15.0 ml of N/2 alcoholic potassium hydroxide. For a blank add to a 250-ml flask (alkali resistant glass), 50 ml of neutralized isopropanol and toluol reagent, 15.0 ml N/2 alcoholic potassium hydroxide and 5 to 15 drops of phenolphthalein indicator solution. Reflux

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the mixture and the blank for 4 hours and back-titrate the excess alkali with N/2 aqueous hydrochloric acid until the reddish tinge disappears. Then boil both solutions vigorously for several minutes to be certain that the reddish tinge does not re-appear.

Calculation of ester number.

$$\text{Ester number} = \frac{(B-A)\ N \times 56.1}{C}$$

where:

A = Milliliters of hydrochloric acid required for titration of the sample.
B = Milliliters of hydrochloric acid required for titration of the blank.
N = Normality of the hydrochloric acid.
C = Grams of sample used.

Calculation of saponification number.

Saponification number = Acid number + ester number.

4.10 Iodine number. The iodine number of the wax shall be determined in accordance with ASTM D 1959.

4.11 Refractive index. The refractive index of the wax shall be determined in accordance with ASTM D 1747.

4.12 Unsaponifiable matter. The unsaponifiable matter in the wax shall be determined in accordance with modified ASTM D 1965.

4.13 Paraffinic hydrocarbon. Oven dry alumina for at least 8 hours at 163 to 177°C (325 to 350°F). Remove from oven and weight immediately. Place in column and cover with heptane. During percolation, maintain the temperature of the solution and the column at the boiling point of the heptane. Determine the paraffinic hydrocarbon in wax in accordance with ASTM D 1342.

4.14 Resinous matter. Weigh out 15 ± 0.1 g of ground wax from the filtered wax composite (ground and sifted by a mechanical sieve shaker to pass through a 40 mesh sieve and to be retained on a 60 mesh sieve) in a 250-ml extraction flask and add 150 ml of acetone at 15° ± 0.5 °C. Maintain temperature and continuously stir for 2 hours. Stir at approximately 300 rpm with 1/4 X 1/2 inch flat stirrer. Keep flask covered to avoid evaporation. After two hours agitation, rapidly filter through a folded filter paper. Of the clear filtrate, collect a 100 ml ± 1 ml aliquot and transfer to a 250 ml tared beaker. Evaporate the solvent on a steam or hot water bath and dry the beaker and extract in an oven at 105°C to constant weight. Calculate the weight in grams of dried extract, multiplied by 10, which is the percentage of cold acetone soluble material.

4.15 Benzene solubility. Weigh out 15 g ± 0.1 g of ground wax from the filtered wax composite (ground and sifted to pass through 20 mesh sieve and be retained on 40 mesh sieve) in a 250-ml extration flask and add 150 ml of benzene c.p. Maintain at a temperature of 25°C for 2 hours, stirring frequently at approximately 300 rpm with a 1/4 X 1/2 inch flat stirrer and keeping the flask covered with a watch glass between stirrings. After 2 hours, filter through a folded filter paper into a 250-ml. Rinse twice with 20 ml portions of benzene and let stand until all the filtrate has flowed into the beaker. Evaporate the benzene of the clear filtrate on steam bath and dry the residue to constant weight in an oven at 105°C.

4.16 Insoluble impurities.

4.16.1 Apparatus.

(a) Condenser, 300 mm.
(b) Dean-Stark moisture trap, 10 ml.
(c) Adapter.
(d) Large mouth round bottom flask, 1000 ml.
(e) Paper extraction thimble, 33/80 mm.
(f) Siphon cup to hold 33/80 mm thimble.
(g) Cotton or glass wool.
(h) Toluol, technical grade.
(i) Xylol, technical grade.

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4.16.2 Apparatus. A suitable assembly of the apparatus is illustrated in figure 1.

4.16.3 Thimble. Place a wad of cotton in the bottom of the thimble, then place a similar wad of cotton in the top of the thimble. Dry the prepared thimble in an oven at 105° to 120°C for 4 to 6 hours. Immediately place in a desiccator until cool or for 1 to 2 hours. Weigh thimble as rapidly as possible and then place in desiccator until ready for use.

4.16.4 Solvent. Pour into flask 180-240 ml of toluol and xylol (1:1).

4.16.5 Sample. Weigh out a 10-gram sample from original sample and place in thimble immediately. First, remove the top wad of cotton, place ground wax in the thimble and cover the wax with the cotton wad.

4.16.6 Procedure. Place the thimble in siphon cup and connect cup to bottom of Dean-Stark tube by means of a wire bow, and assemble apparatus as in figure 1. Boil until all soluble material is extracted. This can be determined by the color of the solvent in the siphon tube of the cup. When the extraction is complete, cool, then remove the thimble from the cup and drain the solvent from the thimble. Dry the thimble in an oven at 105° to 120°C to constant weight and determine the weight of the residue.

$$\text{Percent insoluble impurities} = \frac{\text{Weight of residue} \times 100}{\text{Weight of sample}}$$

4.17 Volatile matter. An aluminum or porcelain evaporating dish (6 cm diameter X 1.8 to 2.2 cm deep) is heated in an oven for 2 hours at 100° to 110°C to constant weight. Weigh out 10 g ± 1.0 mg of wax as specified in 4.4.1 and place in the cooled evaporating dish. Then heat the dish, containing the wax sample, in an oven at 100 to 110°C for 6 to 10 hours or overnight. Remove the dish and wax sample and immediately place in a well-sealed desiccator (with a desiccant of calcium chloride or silica gel) for 1 to 1-1/2 hours until cool. The dish and contents are then reweighed and the percent volatile matter calculated.

$$\text{Percent volatile matter} = \frac{\text{Weight loss on heating} \times 100}{\text{Weight of sample}}$$

4.18 Color. Color standards may be obtained from the American Wax Importers and Refiners Association, Inc. The color of carnauba wax will range from pale yellow to dark yellow for the refined grades and dark yellow to greenish brown for the crude grades.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 All levels. Carnauba wax shall be packed in bags conforming to PPP-B-35, table XIX. The wax, in quantity specified (see 6.2), shall be packed to assure carrier acceptance and safe delivery to destination at lowest rates, in containers complying with the Uniform Freight Classification Rules or National Freight Classification, as applicable.

5.2 Marking.

5.2.1 Civil agencies. In addition to any special markings required by the contract or order, each bag shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to any special markings required by the contract or order, each bag shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Carnauba wax is intended to be used in polishes, finishes, cosmetics, lubricants, inks, protective coatings, medicinals, and as a food additive.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Quantity required.

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(c) Type of wax required (see 1.2).
(d) Color of wax required (see 3.2).
(e) Documentation required.
(f) Special markings required (see 5.2.1 and 5.2.2).
(g) Registration (see 3.3).

6.3 The Brazilian designations for the material covered by this specification are as follows:

Designation Brazilian		American Designation
Type I.		
Grade A --	Flor	-- Flor
--	Primeira	-- Prime or No. 1 Yellow
Grade B --	Mediana	-- Medium or No. 2 Yellow
Type II.		
Grade A --	Cauipe	-- Cauhype
--	Gorda Clara	-- Gorda Clara Light Fatty No. 2 North Country
Grade B --	Gordarosa	-- Fatty Grey or No. 3 North Country

6.4 New classification changes. The following classification changes have been made:

Old designation		New designation	
Grade A --	Yellow types, which include the following:	Type I.	Yellow, which includes the following:
Type 1 --	Flor. Prime or No. 1 Yellow.	Grade A --	Flor. Prime or No. 1 Yellow.
Type 2 --	Medium or No. 2 Yellow.	Grade B --	Medium or No. 2 Yellow.
Type 2A --	Roxa or No. 3 Yellow.	(Deleted).	
Grade B --	Fatty types, which include the following:	Type II.	Fatty, which includes the following:
Type 3 --	Cauhype. -- Gorda Clara. -- Light Fatty -- No. 2 North Country.	Grade A --	Cauhype. -- Gorda Clara. -- Light Fatty or No. 2 North Country.
Type 4 --	Fatty Grey or No. 3 North Country.	Grade B --	Fatty Grey or No. 3 North Country.

MILITARY INTEREST:

The Department of Defense has waived coordination of this document until further notice.

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
CPSC

PREPARING ACTIVITY: GSA - FSS

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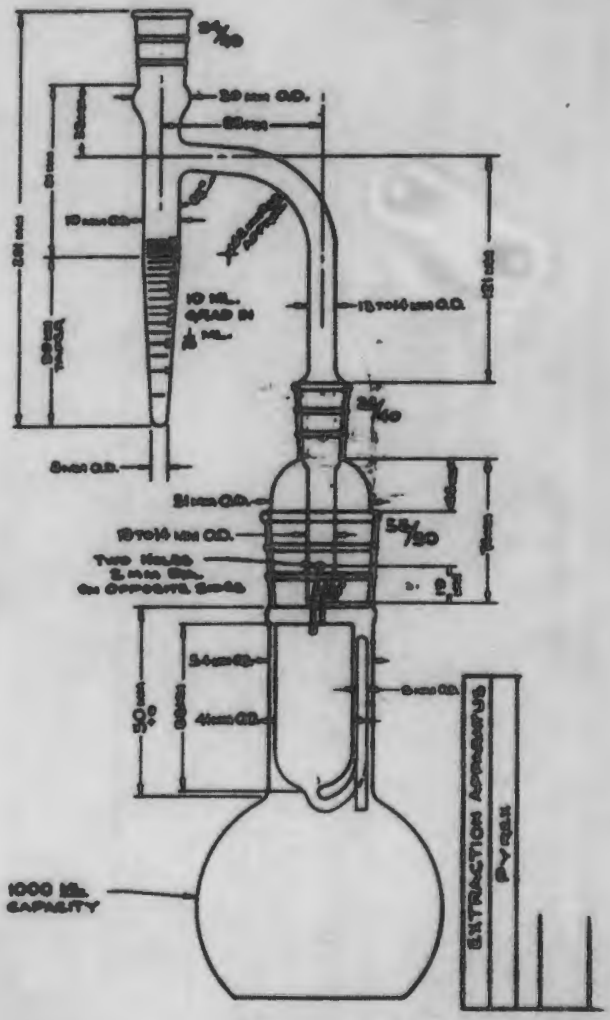


Figure 1. Apparatus for determination of moisture and insoluble impurities.

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